

ENRIQUE GÓMEZ

Biomedical Engineer | Scientific Python Developer | Machine Learning

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Biomedical Engineer and Scientific Python Developer with over three years of experience in scientific software, biosignals, embedded systems, and hardware-software integration. Skilled in developing reproducible Python solutions for signal acquisition and processing, machine learning, optimization, and numerical methods, with experience in cross-validation, statistical analysis, automated testing, technical documentation, and publishing libraries on PyPI. Currently pursuing a Master of Science focused on Artificial Intelligence and researching automated sleep stage classification using EEG signals, graph neural networks, and copula functions.

CORE COMPETENCIES

Programming & Data: Python, NumPy, SciPy, Pandas, Scikit-learn, MLX, SQL, MATLAB, C++, and Java.

Machine Learning: supervised classification, MLP, SVM, PCA, stratified cross-validation, performance metrics, hypothesis testing, and bootstrap methods.

Scientific Software: modular architecture, object-oriented programming, Pytest, unit/integration/robustness testing, Git/GitHub, pyproject.toml, documentation, packaging, and PyPI.

Biomedical & Embedded Systems: EMG, cardiac signals, digital filters, bioinstrumentation, PySerial, USART, sockets, multithreading, sensors, radio frequency, and actuators.

Interfaces & Visualization: Tkinter, PyGame, OpenCV, Matplotlib, and databases.

PROFESSIONAL EXPERIENCE

Associate Professor and Researcher | Universidad Autónoma de Aguascalientes · CCI/CCB | 2023 - Present

- Teach courses in Python, programming, electronics, embedded systems, bioinstrumentation, and biomedical measurements.
- Design and implement hands-on activities focused on programming, robotics, applied electronics, and biomedical solutions.
- Supervise academic projects centered on technological innovation, problem-solving, and multidisciplinary collaboration.

Instructor | Universidad Cuauhtémoc · School of Medicine | 2024 - 2025

Research Methodology I-II

- Provided technical guidance for research protocols, preliminary proposals, and thesis structures.
- Advised on experimental design, validation methodology selection, scientific writing, and specialized literature analysis.

Python and Embedded Systems Developer | TomorrowLabs | 2023 - 2024

- Developed an interface and API for digitizing and visualizing EMG signals via USART.
- Implemented database acquisition, CRUD operations, multithreaded processes, and integration of sensors, radio-frequency modules, and actuators.

Research Intern | CIBION · Universidad Autónoma de Aguascalientes | 2022

- Contributed to a device for cardiac signal acquisition, digital filtering, biomedical visualization, and technology planning.

FEATURED PROJECTS & SCIENTIFIC SOFTWARE

Sleep Stage Classification with GNNs and Copula Functions - Master's Thesis Project 2025 - Present | UAA

- Developing a methodology for automated sleep stage classification from EEG signals using graph neural networks enriched with copula functions.
- Modeling functional brain connectivity and multivariate nonlinear dependencies, with planned evaluation across Sleep-EDF, MASS, and ISRUC-Sleep datasets.
- Building a reproducible preprocessing, training, and validation pipeline, and exploring matrix acceleration with MLX on Apple Silicon.

EEEE-Py - Technical Lead and Developer

2026 | [PyPI](#)

- Led the modular architecture, repository, and testing of a library implementing EES, EMNA, PSO, and GWO; developed a 30-run Python-versus-R benchmark against its R counterpart.
- Confirmed statistically equivalent solution quality (Mann-Whitney U, $p = 0.8418$) and approximately 2.5 times faster execution in Python.

EvoBench - Lead Author and Developer

2026 | [PyPI](#)

- Led the architecture, implementation, testing, documentation, and publication; integrated EDA, PSO, ABC, benchmark functions, an experimental engine, and statistical analysis.

MNIST Bench - Author and Developer

2026

- Designed a reproducible 10-fold stratified cross-validation pipeline to compare an MLP and a one-vs-one SVC.
- Achieved 96.72% accuracy with MLP versus 88.16% with SVC; validated the difference using a paired t-test ($p < 0.001$), Wilcoxon testing, and bootstrap analysis at the 95% confidence level.

Pixiter - Co-developer

2025

- Compared global Lagrange interpolation, local planar interpolation, and random filling; the local approach reduced average grayscale error from 101.2% to 34.69% and demonstrated greater numerical stability.

EDUCATION

Master of Science in Computing and Applied Mathematics | Universidad Autónoma de Aguascalientes | 2025 - Present

Focus: Artificial Intelligence

Thesis in progress: Sleep Stage Classification Using a Graph Neural Network Architecture Enriched with Copula Functions.

Bachelor of Science in Biomedical Engineering | Universidad Autónoma de Aguascalientes | 2017 - 2022

LANGUAGES

Spanish: Native. English: B2. Mandarin Chinese: Preparing for HSK Level 1 certification.